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NSP/004/106 - (OHI 6) Guidance on the selection and application of conductor joints, terminations & binders

1. Purpose

The purpose of this document is to provide guidance on the selection and application of conductor joints, terminations & helical fittings for use on wood pole lines for use on the CE Electric UK Distribution System.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/106	1 (May 06)	(OHI 6) Guidance on the selection and application of conductor joints, terminations & binders
OHI 6	Feb 1992	Conductor Joints, Terminations and Binders
RTN / 001 / 500 / 711 / 006	Feb 1997	GN 71-06 Guidance note for Helical preformed fittings
RTN / 001 / 500 / 711 / 001	Sept 1995	GN 71-01 Guidance note for Compression Joint Techniques

2. Scope

The information provided in this guidance document is limited to full tension joints and all types of helical fitting for use on wood pole overhead lines up to and including 33kV. Details on Non-tension joints and jumpers have been provided in NSP/004/107.

3. Technical Specification

3.1 Full tension joints

Conductor joints are manufactured to comply with NPS/001/016. Generally a conductor joint must have conductivity characteristics comparable with that of the associated conductor, be capable of carrying the full line current without overheating and also short duration maximum fault current of the line without joint failure. In line with the above specifications, mechanically all joints are required to withstand 95% of the UTS of the associated conductor. As a result certain conductor sizes require the application of 'Hex press' type fittings even though 'Versa Press' fittings may physically fit the conductor size.

Full tension joints shall be provided through the application of one of the following types of compression jointing systems.

- a) 'Versa Press'
- b) 'Hex Press'

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Full tension joints

Conductor Type / Size	Compression system	Selection chart drawing number
HDBC & Ca CU	Versa press	1091010102 Sheet 2
50-100mm AAAC	Versa press	1091010102 sheet 3
50 -100mm ACSR	Versa press	
150mm ACSR (18+1/3.35mm) (Dingo)	Versa press	
70 – 175mm ACSR	Hex press	1091010653 sheet 1
175mm AAAC	Hex press	1091010653 sheet 2
200mm AAAC	Hex press	TBA

3.2 Armour Splice

Armour Splices are a range of helical fittings which can be applied over the outer strands of a damaged conductor. (See NPS/001/002 for specification information). Once applied they will provide the conductor back to its original UTS. They are approved for use on hard drawn bare copper, copper cadmium, aluminium alloy and ACSR conductors, as shown on drawing no 1091010657.

They shall only be used as a repair medium when one third or less of the outer strands have suffered damage.

Damage to the steel core of ACSR conductors must be repaired with the appropriate type of full tension compression joints.

3.3 Terminations

Where a choice of conductor termination method is available, the preferred arrangement will be to terminate conductors using preformed helical deadends. The tail from the deadend will then be carried through with a lugged connection or extended with a non tension connector.

Compression fittings will normally only be used in the following situations:

- Installation of a live line section point where the conductors are live. (See drawing 1091010573 for details on the preferred arrangements for creating live line sections).
- Large CSA conductors on EHV Lines
- ACSR conductors and Covered conductors
- At jumpers positions in main line routes where a live line disconnect facility is unlikely to be required.

3.3.1 LV Lines

Conductor Type / Size	Type of Fitting	Selection chart drawing number
Main Lines		
Open Wire Construction	Helical Deadend	1091193155 sheet 1
ABC Conductor – Full Tension 2 Core & 4 Core	Anchor Clamp	1000431410 sheet 1
Services		
Hybrid Concentric	Helical Deadend	1091193152 sheet1
ABC Conductor – Reduced Tension 2 Core & 4 Core	Anchor Clamp	1000431410 sheet 1
16mm Open Wire Service Aerial (Historical)	Helical Deadend	1091193155 sheet 1

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3.3.2 HV Lines (11 – 33kV)

Conductor Type / Size	Type of Fitting	Selection chart drawing number
All conductor types (see CC below)	Helical Deadend	1091010425
Or		
All Conductor Types (see CC below)	Compression Dead End	1091010102 sheet 9
Covered Conductor	Compression dead end	1091010667 sheet 1

3.3.3 EHV Lines (33kV – 66kV)

Conductor Type / Size	Type of Fitting	Selection chart drawing number
All conductor types	Helical Deadend	1091010425
Or		
100mm AAAC (Oak), 100mm HDBC, 125mm HDBC	Compression Dead End	1091010102 sheet 9
175mm AAAC (Elm), 200mm AAAC (Poplar)		1091010653 sheet 2
70mm ACSR (Horse), 150mm ACSR (Wolf), 175mm ACSR (Lynx)	Compression Dead End	1091010653 sheet 1

3.4 Intermediate or Angle Binder/Ties

Helical preformed fittings are essentially basically groups of conductive rods formed into a helix shape giving the appearance of a stretched spring. The inner diameter of the helix is slightly smaller than that of the conductor or stand to which the fitting will be applied giving the effect of a high gripping force without causing potentially damaging stress to the conductor or strand. Helical preformed fittings can also be used to attach stays to poles. See NPS/001/002 for specification details

3.4.1 LV Lines

Conductor Type / Size	Type of Fitting	Selection chart drawing number
Main Lines		
Open Wire Construction (intermediate/angle)	Helical Fitting	1091193156 sheet 1
ABC Conductor – Full Tension 2 Core & 4 Core	Roller suspension clamps	1000431406 sheet 2
Services Lines		
ABC Conductor – Reduced Tension 2 Core & 4 Core	Roller suspension clamps	1000431406 sheet 2

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3.4.2 HV Lines (11 – 33kV)

Selection of fittings:

For ease of identification of the strand or conductor range accommodated by an individual preformed fitting a colour code paint mark is applied to the fittings.

- a) Use top ties on all line types to secure conductors to pin insulators at in-line situations in normal environments.
- b) Use side ties on all line types to secure conductors to pin insulators at angle poles in both normal and high altitudes
- c) Use double top ties to secure large conductor types to pin insulators at in-line situations in high altitude environments or where difficulty is encountered when securing large conductors e.g. down steep inclines.
- d) Use twin grip uplift ties to secure small conductor types to pin insulators where lines are constructed down steep inclines or where small levels of uplift exist in normal environments only.

Conductor Type / Size	Type of Fitting	Selection chart drawing number
(Intermediate Top Ties) - All conductor types including covered conductor	Helical Top Tie	1091010660
(Double Top Ties) - All conductor types excluding covered conductor	Helical Double Top Ties	TBA
Twin grip Uplift Ties (Limited to lines with CSA of <=50mm AAAC or 32mm HDBC)	Helical Uplift Tie	1091010142
Intermediate Ties – Covered conductor	Helical Top Tie	1091010660
Conductor Type / Size	Type of Fitting	Selection chart drawing number
(Angle Ties) – All conductor types including covered conductor	Helical Side Tie	1091010662

Notes

- 1) Where ties are required pilot pin insulators, they shall be installed using the following types
 - a) Straight jumpers 1091010660
 - b) Angled jumpers 1091010662

3.4.3 EHV Lines (33kV – 66kV)

Conductor Type / Size	Type of Fitting	Selection chart drawing number
All conductor types	Helical Side Tie	1091010662

3.5 Helical fittings for stays

See NSP/004/104 for further details on approved fittings types for terminating stays wire

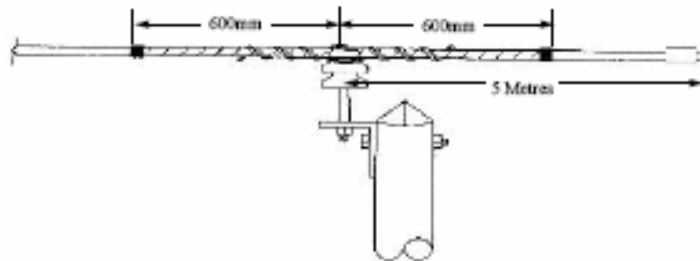
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3.6 Application of helical fittings – general

- When applying preformed fittings, the starting point of application on dead ends and helical ties is easily determined by paint markings applied to the fittings at the beginning of the helix.
- Application instructions for each type of helical fitting can be found in the following DBD related documents WI 71-01, WI 71-02, WI 71-03, WI 71-04, WI 71-05, WI 71-10, WI 71-11, WI 71-12, WI 71-13.

3.7 Application of helical fittings onto covered conductors

- To overcome television interference caused by electrical discharges at the insulators, the XLPE or PVC covered conductor shall be wrapped with semi-conducting tape for 600mm each side of the insulator centre before binding in the conductor. Note The tie pads are not required in these circumstances. PVC covered conductor shall no longer be installed, however where it is retained the following additional info shall apply.
- At existing PVC insulated positions make sure that the distance between the insulator and the point where the PVC covering is terminated is not less than 5m.
- Where XLPE or PVC coverings are removed from the conductor, the ends of the conductor or the exposed part on XLPE covered lines shall be insulated to prevent longitudinal moisture ingress and ore restore the electrical properties of the insulation. This shall be achieved by the application of a 50% overlapped layer of self amalgamating tape. See WI 71-04 Sealing PVC covered conductors



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4. References

4.1 External Documentation

Reference	Title
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4.2 Internal documentation

Reference	Title
NPS/001/016	Technical specification for compression fittings and mechanical conductor fittings
NSP/004/107	OHI 7 Guidance on the selection of conductor Jumpers and Non-tension connections on wood pole lines
NSP/004/104	OHI 4 Guidance on the types and installation requirements for stays
NPS/001/002	Technical specification for helical fittings
WI 71 -01	Application of helical Top Ties
WI 71 -02	Application of Helical Double Top Ties
WI 71 -03	Application of Helical Side Ties
WI 71 -04	Sealing of PVC Covered Conductors
WI 71 -05	Application of Helical Twin Grip ties
WI 71 -10	Application of Helical Dead end Stay Grip to a stay rod
WI 71 -11	Application of Helical Dead end Stay Grip to a stay insulator
WI 71 -12	Application of a Helical Stay Splice
WI 71 -13	Application of a Helical Pole top wrap
WI 71 -14	Application of a Helical Insulator Link Assembly
OHI 4	Application of Stays and stay Fittings

5. Definitions

Term	Definition
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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	04/01/07

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	08/01/07

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	03/01/07

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South	D Sillito	29/01/07
S Gardner	OHL Construction Manager	S Gardner	23/03/07

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
Neil Denley	Primary Engineering Projects Manager	Neil Denley	29/03/07